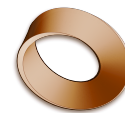


Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics



Planning an Investigation

Homework 1 · Criterion B practice

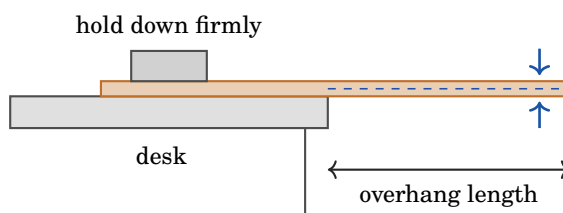
Due: Thursday 24 September

 **About one hour.**

You are about to plan and carry out the **pendulum investigation** in class, and it is a **summative** — so every word of it is written in the lesson, in your own words, from your own data. **None of it goes home.**

So this packet practises the *planning* — **Criterion B** — on somebody else's experiment. Get good at the writing here, and when you sit down in class with your own investigation the writing is the easy part. **Keep your Criteria B and C sheet beside you: each section below is one row of it.**

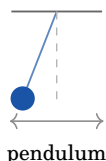
Meet the experiment. Kit and Sam hold a metre rule flat on a desk with part of it over the edge, press the desk end down hard, and twang the overhanging end. It bounces up and down — it **oscillates**. They want to know whether the **length of the overhang** changes **how fast it oscillates**.



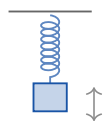
Reading — oscillations and frequency

An **oscillation** is a repeating back-and-forth movement about a middle position. The middle is where the object sits when nothing disturbs it, and an oscillating object keeps passing back through it.

You have met more oscillations today than you realise — every one of them the same idea wearing a different coat.



pendulum



mass on a spring



twanged ruler

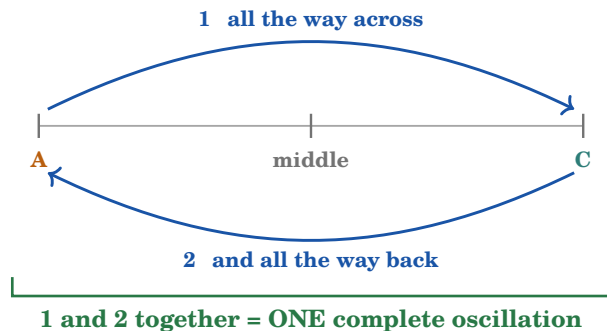
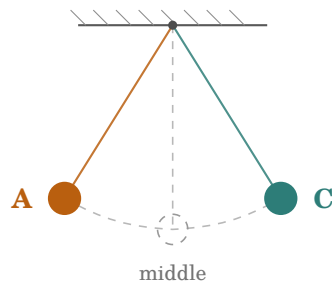


guitar string



your vocal cords

Counting one oscillation — where nearly everyone goes wrong

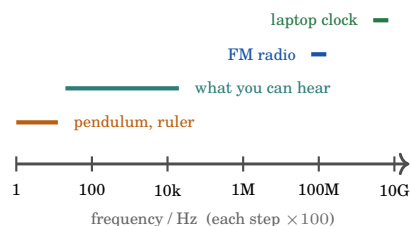


A pendulum that swings from A across to C has done **half** an oscillation. It is not finished until it is back at A doing exactly what it was doing at the start.

Frequency, and the hertz

The **frequency** of an oscillation is **how many complete oscillations happen in one second**, measured in **hertz (Hz)**. One hertz is one oscillation per second.

$$f = \frac{\text{number of oscillations}}{\text{time taken}}$$



Because real frequencies get very large, we use prefixes: **kHz** is a thousand hertz, **MHz** a million, **GHz** a thousand million. The scale opposite shows how far apart they really are — each step along it is a hundred times faster than the last.

Worked example. A twanged ruler completes 30 oscillations in 6 seconds.

$$f = \frac{\text{oscillations}}{\text{time}} = \frac{30}{6} = \mathbf{5 \text{ Hz}}$$

The answer is *per second*, so the unit is hertz. Always write the unit.

1. A pendulum completes 20 oscillations in 40 s. $f = \mathbf{0.5 \text{ Hz}}$
2. A ruler completes 45 oscillations in 5 s. $f = \mathbf{9 \text{ Hz}}$
3. Write 3 kHz in hertz $\mathbf{3000 \text{ Hz}}$ and 2 500 000 Hz with a prefix $\mathbf{2.5 \text{ MHz}}$
4. A bee's wings beat 12 000 times in one minute. Show that this is 200 Hz.

12 000 beats happen in 60 seconds.

12 000 ÷ 60 = 200 beats every second, and that is 200 Hz.

Marking: The minute is the trap — frequency is always per second, so turn one minute into 60 s before dividing.

CRITERION B • strand i The question

A good scientific question is **testable**: a reader can tell, without asking you, **what you will change** and **what you will measure**.

How does what you CHANGE affect what you MEASURE ?

5. Kit wrote three possible questions. **Tick** the one that is testable, and say what is wrong with the other two.

✓	Kit's question	What is wrong with it?
	Is a long ruler better than a short one?	"Better" is not something you can measure.
✓	How does the length of the overhang affect the frequency of the ruler?	Nothing — it says what changes and what you measure.
	Why do rulers oscillate?	Nothing gets changed, so there is no experiment to do.

6. Rewrite this so it is testable: *"Does the size of a drum change the sound?"*

How does the width of a drum skin affect the frequency of the note it makes?

Marking: Accept any question that names one thing being changed and one thing being measured.

CRITERION B • strand ii The hypothesis

A **hypothesis** is not a guess — it is a **prediction with a reason attached**, and the reason must be physics.

I think _____ will _____

the prediction

because _____



the marks are here

7. Kit wrote: *"I think a shorter overhang will oscillate faster."* That is a prediction and nothing more — on the criteria sheet it sits in the bottom band. Add a reason, starting with *because*.

...because a shorter piece of ruler is stiffer, and there is less of it

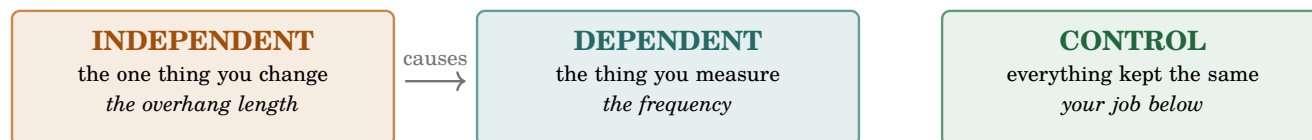
moving. It springs back to the middle faster, so it fits more

oscillations into every second.

Marking: Accept stiffness, length, or the amount of ruler that is moving, as the reason. "Because it is smaller" is not enough — it just says the change again instead of explaining why that makes a difference.

CRITERION B • strand iii Variables, and the data you will collect

One thing changed, one thing measured, everything else held still.



8. Name **three** things Kit must keep the same, and say what would go wrong if each one changed. Then answer the question underneath — the criteria sheet asks for **sufficient** data, not just data.

	Kept the same	What would go wrong if it changed
1	<i>the same ruler</i>	<i>a different ruler bends differently, so the frequency would change for a reason that is not the overhang</i>
2	<i>how hard it is twanged</i>	<i>a harder twang makes a bigger swing, which is harder to time the same way each time</i>
3	<i>how firmly the desk end is held</i>	<i>if the hold is loose, more of the ruler moves than the bit you measured</i>

Kit plans to test **two** overhang lengths, **once each**. Why is that not enough, and what should Kit do instead?

Two points always make a straight line, so you cannot tell what the real pattern is. With only one go at each length, you would never spot a reading that had gone wrong. Kit should try five lengths, three times each, and average.

CRITERION B • strand iv The method

Numbered steps a classmate could follow **without asking you a single question**, plus equipment and one safety point.

9. Kit's method has **four** faults. Write what is wrong beside each step.

Kit's method	What is wrong with this step?
1. Put the ruler on the desk.	It does not say how far the ruler should stick out over the edge — and that is the thing being changed.
2. Twang it.	It does not say how hard to twang it, so it is not kept the same each time.
3. Time it with a stopwatch.	Time what? It does not say how many oscillations to count, and it does not give a unit.
4. Write down the answer.	There are no repeats, and it never says to change the overhang and try again.

10. Now write the method properly — five or six numbered steps. A classmate must be able to follow it without asking you anything.

1. Put the metre rule flat on the desk with exactly 80 cm over the edge.

2. Hold the desk end down hard so it cannot lift.

3. Pull the free end down 5 cm and let go.

4. Start the stopwatch as you let go, and time 10 full oscillations.

5. Write the time down. Do the same two more times at 80 cm.

6. Move the rule to 70 cm and do it all again.

7. Repeat for 60 cm, 50 cm and 40 cm.

Marking: Credit a measured overhang, a controlled release, a stated number of oscillations, repeats at each length, and a range of lengths.

11. A method is not finished until a reader knows what to fetch and what to watch out for. Complete both boxes.

- **Equipment** — everything a classmate would have to collect *before* starting. Be specific: *a metre rule*, not *a ruler*. Remember Kit has to measure the overhang as well as time it. One item per line.
- **Safety point** — name something that could genuinely go wrong in *this* experiment, and say what you would do about it. **“Be careful” is not a safety point** — it names no hazard and changes nothing.

Equipment	Safety point — the hazard, and what you would do
metre rule	A twanged rule can spring off the desk and
a second rule, to measure	hit someone's hand or eye. Keep faces away
the overhang	from the free end, do not bend it too far, and
stopwatch	hold the desk end down hard so it cannot fly up.
a desk with a straight edge	

Before you hand this in

Criterion B is written **before you touch the apparatus**. Tick each row.

✓	Strand	Have you...
	B i	...written a question naming what you change <i>and</i> what you measure?
	B ii	...predicted an outcome <i>and</i> given a physics reason after the word <i>because</i> ?
	B iii	...named the independent, the dependent and at least three controls — and said how many readings and how many repeats?
	B iv	...written numbered steps a classmate could follow unaided, with equipment and one safety point?

The five words that decide your level

Your criteria sheet uses the same handful of verbs over and over. The strand barely changes between levels — the *verb* does. Learn these five and you can aim at a band deliberately instead of hoping.

states	you wrote it down
outlines	you gave the short version
describes	you said what it is really like
explains	you said <i>why</i>
discusses	you weighed it up, both sides

Words used in this packet

oscillation	a repeating back-and-forth movement about a middle position
one oscillation	all the way there <i>and</i> all the way back, to where it started
frequency	how many complete oscillations happen in one second. Symbol <i>f</i> , unit hertz (Hz)
hertz	one oscillation per second. kHz = 1 000 Hz, MHz = 1 000 000 Hz
hypothesis	a prediction <i>with a scientific reason attached</i>
independent	the one thing you deliberately change
dependent	the thing you measure
control	anything you keep the same so it cannot affect the result
repeat	doing the same measurement again at the same setting